31_DataVisualization

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```r
source("01_datacleaning.R")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'janitor'
##
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
##
##
## Rows: 144 Columns: 22
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): chamber
## dbl (21): congress, year, party.mean.diff.d1, prop.moderate.d1, prop.moderat...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 774 Columns: 25
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (24): Candidate, State, District, Office Type, Race Type, Race Primary E...
## dbl  (1): Primary %
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)

## Rows: 1599 Columns: 27
```

```
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (26): Candidate, Gender, Race 1, Race 2, Incumbent, Incumbent Challenger...
## lgl  (1): Race 3
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 811 Columns: 32
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (30): Candidate, State, District, Office Type, Race Type, Race Primary E...
## dbl  (2): Partisan Lean, Primary %
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 1077 Columns: 26
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (26): Candidate, Gender, Race 1, Race 2, Race 3, Incumbent, Incumbent Ch...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

TITLE: How Demographics and Party Dynamics Shape Primaries

ABSTRACT:

This project explores how party dynamics and candidate demographics relate to primary election success in the United States. Using statewide Democratic and Republican primary candidate data from the 2022 election cycle, we examine the relationships between gender, race, party, and office type in shaping who wins and who loses. Our visualizations show that gender on its own is not a strong or consistent predictor of success, but once the data is separated by party and office type, clear differences begin to emerge.

Women tend to perform better than men in Democratic primaries, while men continue to dominate Republican primaries, especially in more competitive Senate contests. Race also interacts with these patterns, and some gender–race combinations experience stronger or weaker outcomes depending on the political environment. Our logistic regression results confirm that primary vote percentage is the single strongest factor in determining who wins a primary, but even after controlling for vote share, party and race still show meaningful associations with outcomes.

A secondary question on ideology changes will be further explored to understand how differences in ideological positioning relate to these structural patterns. We also look at ideology to see whether being more moderate or more extreme affects a candidate's chances of winning. Early patterns show that ideology might matter differently for different groups. For example, some gender–race combinations seem to do better when they are closer to the party's mainstream, while others face more challenges when they take more ideological positions.

INTRO:

The purpose of this study is to understand how gender, race, and ideological placement relate to political success in U.S. primary elections and to see how these patterns differ between Democrats and Republicans. Primary elections play a major role in determining who appears on the general election ballot, so studying who wins primaries can tell us a lot about representation, competitiveness, and the types of candidates each party tends to reward.

The first research question we explore asks how gender correlates with primary win rates and whether this relationship is different across parties or office types such as Governor and Senate. Past research suggests that gender and race can influence electoral outcomes, but these effects often depend heavily on the political context, making it important to examine Democrats and Republicans separately. The second research question, which will be addressed later, will focus on ideology by looking at how ideological differences between parties relate to broader dynamics in primary competitiveness. Throughout the report, we use a mix of visualizations and logistic regression models to identify where demographic traits matter most and where larger structural patterns shape election outcomes.

The second research question we explore looks at ideology and how the ideological positions of the two major parties have changed over time. To examine this, we use two visualizations: one that shows how much the ideological distributions of Democrats and Republicans overlap from year to year, and another that tracks the growing ideological distance between the parties. These graphs help us see whether increasing polarization shapes primary competitiveness and whether candidates are rewarded for aligning more closely with their party's ideological center or its extremes.

DATA:

The dataset consists of all Democratic and Republican statewide primary candidates from the 2022 election cycle. Each row represents a candidate and includes information about the candidate's gender, race, party affiliation, office sought (Senate or Governor), primary vote percentage, and primary outcome. The dataset also includes ideological scores that will be used later in the second part of the project. The 2022 cycle offers a large and balanced sample across both parties, making it a strong foundation for exploring demographic differences in performance and success.

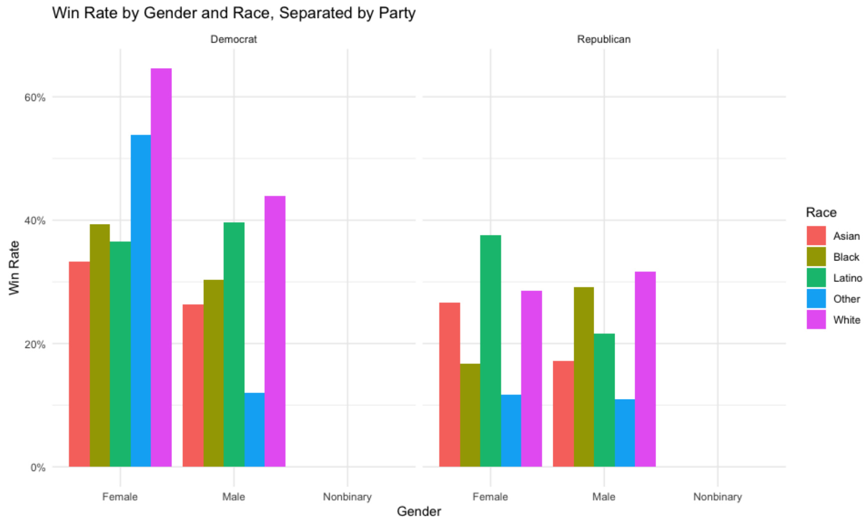
For the second dataset, which tracks ideological polarization in Congress over time, we used the provided file containing yearly ideology scores for Democratic and Republican legislators. This dataset includes measures such as party means, ideological spread, and distance between the parties in each year.

DATA CLEANING:

To prepare the dataset for analysis, we followed a strong cleaning process. Missing or inconsistent demographic information was removed to ensure fair comparisons. Gender labels were standardized into "Female," "Male," and "Nonbinary," and primary election outcomes were standardized into a binary format showing if the candidate won (1) or lost (0). Because this project focuses specifically on the two major political parties, we filtered out minor-party candidates. Primary vote percentages, which appear in different formats in the raw data, were converted into numeric values so they could be used directly in the regression model. We also created grouped data structures that combine race, gender, party, and office type, which allowed us to build heatmaps and intersectional analyses later in the report. These steps ensured that the dataset was clean, consistent, and ready for both visual and statistical modeling.

For the second dataset, we cleaned the data by selecting the variables needed for our two visualizations—overlap between the parties and ideological distance over time standardizing column names, and removing years with incomplete information. We also converted party labels and ideology values into a consistent numeric format so the trends could be plotted clearly. This cleaned dataset allowed us to create the two graphs used in the ideology section and to compare long-term ideological change to the patterns we observed in the primary candidate data.

VISUALIZATION 1: Win Rate by Gender and Party

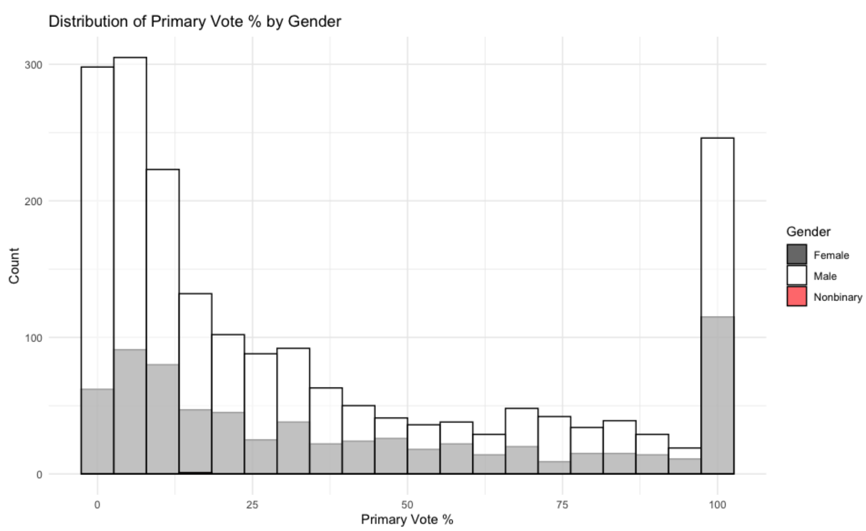


When we look at win rates by gender across the entire dataset, the differences aren't large. However, once we split the results by political party, the patterns become much clearer.

In Democratic primaries, women tend to win at higher rates than men. This could reflect a combination of factors, such as Democratic voters being more willing to support women, a lot more greater institutional encouragement inside the Democratic Party for women to run, or women being more selective and entering races where they know they have stronger chances of success.

In Republican primaries, the trend is the opposite. Men win far more often than women, which may potentially reflect deeper structural differences within the party, such as fewer women being recruited, fewer women receiving financial support, or ideological currents within Republican politics that may make it more difficult for women to gain traction in primaries. The giant contrast between the two parties shows that gender effects are not uniform. Instead, gender interacts heavily with party culture, different recruitment pipelines, the competitiveness of candidate pools, and the types of candidates each party tends to reward.

Visualization 2: Distribution of Primary Vote Percentage by Gender

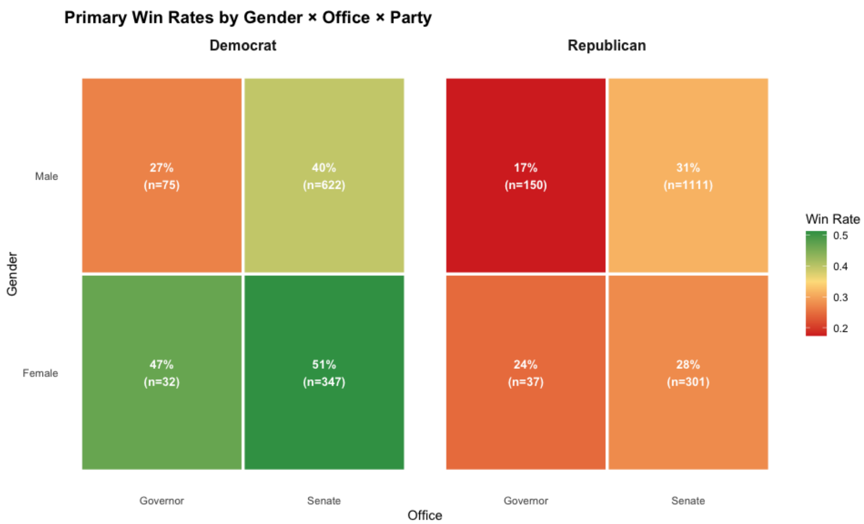


Looking at how primary vote percentages are distributed across genders helps us understand not just who wins but how candidates perform overall. The histogram shows that female candidates tend to cluster more in the mid-to-high range of vote percentages, meaning that even when women don't win, they tend to remain competitive throughout the race.

Male candidates show a much wider spread, including both extremely strong performances and a larger number of poor-performing campaigns. This suggests that men enter a larger variety of races, some of which may be unwinnable, whereas women may be more strategic or may face more structural filtering before entering a race.

Nonbinary candidates appear too infrequently for meaningful comparison, but their limited presence reflects broader patterns in statewide politics. While the differences in distribution are not extreme, they do suggest that women tend to run more consistently strong campaigns overall, while men run in a wider variety of campaign contexts with more variability in performance.

Visualization 3: Primary Win Rate by Race X Gender X Party



Race adds another important layer to understanding primary outcomes.

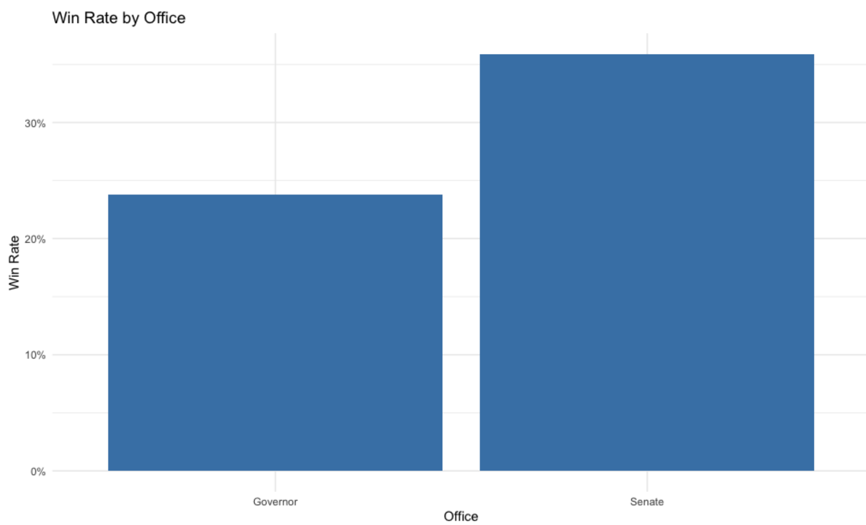
When we examine win rates across racial groups alone, we see that White and Black candidates tend to have higher success rates, Asian candidates fall in the middle, and Latino or "Other" candidates have somewhat lower success rates.

However, these overall averages don't tell the full story because the role of race becomes clearer when we combine it with gender.

In intersectional visualizations, White women tend to perform especially well in Democratic primaries, and Black women also show relatively strong success rates, though with smaller sample sizes. Latino men, on the other hand, show lower win rates across both parties. These patterns become even more pronounced when we separate results by party. In Democratic primaries, racial gaps are smaller, and women of different racial backgrounds win at relatively similar rates.

In Republican primaries, the gaps widen significantly, with White men dominating success rates and most other racial and gender groups experiencing lower outcomes. These findings reflect broader political dynamics in the United States, where the Democratic Party tends to field a more diverse set of candidates and has voters more willing to support racial and gender diversity, while the Republican Party tends to favor candidates who align more closely with its historically White, male base of primary voters.

Visualization 4: Win Rates by Office Type: Senate vs Governor

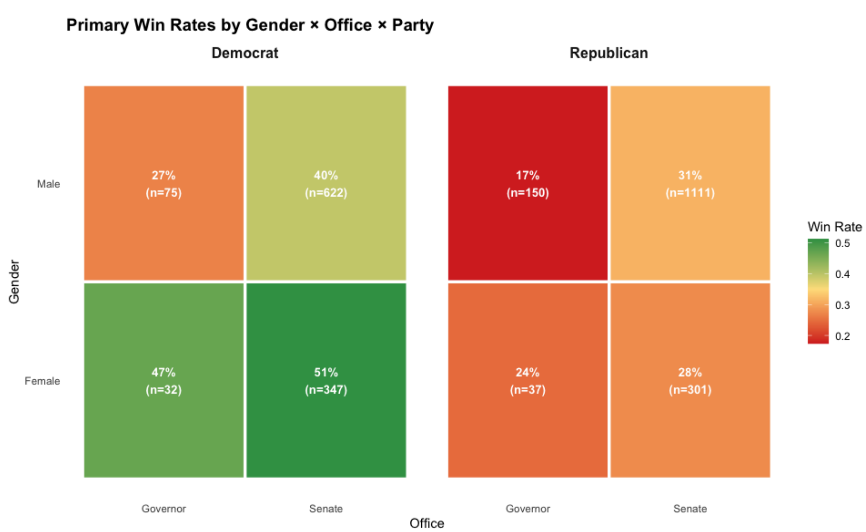


Office type also plays a major role in shaping primary outcomes. Senate races tend to be more competitive because they attract larger pools of candidates, have much higher stakes, and typically involve fewer seats. As a result, the overall win rates in Senate races are lower, simply because candidates face more competition.

Governor races, on the other hand, often have clearer frontrunners, fewer challengers, and more defined party support structures, which leads to higher average win rates. Because women and men do not enter each office type at the same rate, these structural differences can contribute to gender gaps.

If women are more likely to run for Senate seats, where success rates are naturally lower, their overall win rates may appear worse even if they perform just as competitively within their own office category. Understanding office type is important because it highlights how structural competitiveness and not just demographic traits shapes electoral outcomes.

Visualization 5: Heatmap: Gender × Office × Party

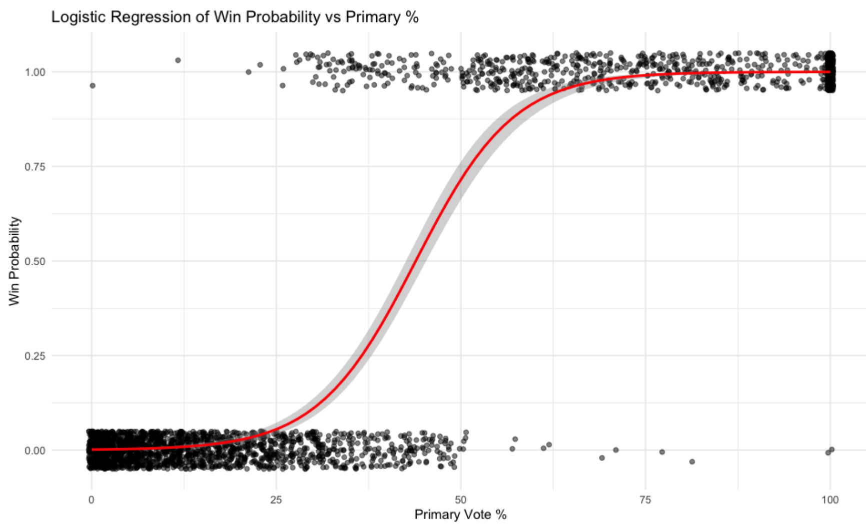


The heatmap brings together gender, office type, and party into a single visualization, and it is one of the most interesting figures in the analysis.

In Democratic primaries, women show consistently strong success rates across both Governor and Senate races. Men also perform well, but women often hold a slight advantage, suggesting that Democratic voters and institutions may be more favorable environments for female candidates.

In Republican primaries, this pattern flips. Women consistently show lower success rates across both offices, but especially in Senate contests, where the gender gap widens substantially. Men continue to dominate across most Republican categories. What makes the heatmap so strong is that it shows how gender differences are not simply "men do better" or "women do better." Instead, the relationship depends on the interaction of three major structural factors: the political environment of the party, the level of competitiveness of the office, and the gender composition of the candidate pool. These combinations help explain the earlier results and clarify why gender alone is not a consistent predictor of success.

Visualization: Logistic Regression: Main Predictors of Winning




```r
knitr::include_graphics("logistic_reg.jpg")
```

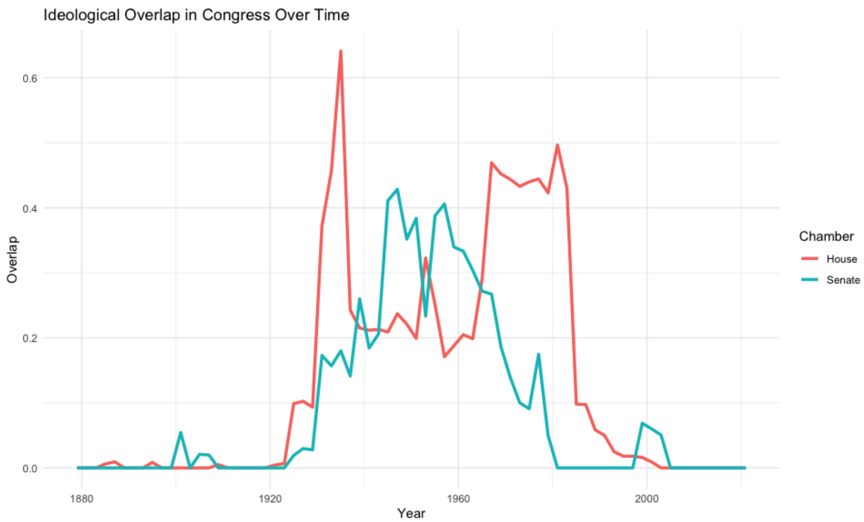
	OR	2.5 %	97.5 %
(Intercept)	0.0002573010	4.467958e-05	1.279775e-03
genderMale	1.0006691458	6.387787e-01	1.576158e+00
genderNonbinary	0.0001426958	NA	3.307716e+43
raceBlack	3.8164490094	9.987987e-01	1.578876e+01
raceLatino	2.0350401875	4.915290e-01	8.953010e+00
raceOther	0.6561307962	9.275983e-02	4.483411e+00
raceWhite	3.9516255769	1.128877e+00	1.517703e+01
partyRepublican	1.9607199075	1.256875e+00	3.098413e+00
officeSenate	0.7418744534	3.734122e-01	1.500918e+00
primary_percent	1.1700868332	1.152721e+00	1.189736e+00

The logistic regression model tests whether gender still matters after controlling for other important predictors such as party, race, office type, and primary vote percentage. The results show that primary vote percentage is by far the strongest predictor of winning, which makes more logical sense because candidates who receive more votes are overwhelmingly the ones who win. In fact, each additional percentage point of vote share significantly increases a candidate's odds of victory. The party also remains important, with Republican candidates showing higher odds of winning, more likely because Republican primaries in this dataset tend to have fewer candidates per race. After controlling for vote share and party, gender itself is not statistically significant, meaning that gender does not independently determine the likelihood of winning.

Race shows some associations with win probability, but the estimates come with uncertainty due to sample size differences. Office type continues to matter as well, with Senate candidates having slightly lower odds relative to Governor candidates, echoing the earlier observation that Senate races are more competitive.

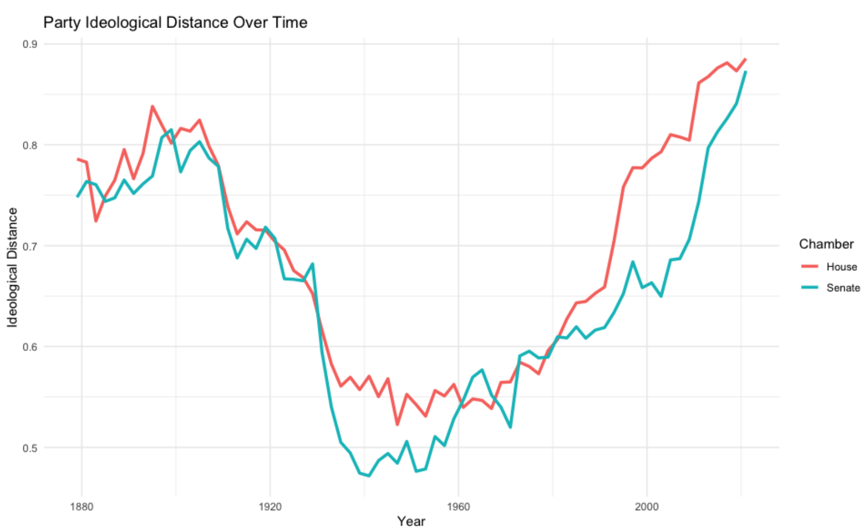
The regression results confirm that the raw gender gaps seen in the visualizations arise mostly from contextual differences such as party environments, competitiveness, and candidate pool composition rather than gender acting as a direct causal factor.

VISUALIZATION: IDEOLOGICAL OVERLAP IN CONGRESS



The first plot flips the perspective by showing overlap—the amount of shared ideological space between Democrats and Republicans. Higher values mean the parties have members who vote similarly; lower values mean there is almost no ideological mixing. Early in U.S. history, overlap was relatively high, especially in moments where conservative Democrats and liberal Republicans existed. Overlap peaked around the 1920s–1940s, but then fell sharply beginning in the 1970s. Today, overlap is essentially zero, meaning the parties no longer share any meaningful ideological common ground. The House loses its overlap slightly faster than the Senate, but both chambers ultimately converge to the same place: near-total separation. This visualization reinforces the story from the first plot — modern Congress is defined by extreme partisan separation, with almost no ideological crossover remaining.

VISUALIZATION:



The second plot shows how far apart Democrats and Republicans are ideologically in the House and Senate from the late 1800s to today. Higher values mean the parties are more polarized — their average ideological positions are farther apart. The patterns show that polarization dropped sharply around the mid-20th century, reaching its lowest point between the 1940s and 1950s, when the parties were the most ideologically similar. After roughly 1970, the distance between the parties begins rising steadily in both chambers, with the modern era showing the highest levels of polarization on record. The House consistently polarizes earlier and more sharply than the Senate. Overall, this visualization makes it clear that Congress has become dramatically more polarized over time, especially since the 1980s.

Conclusion:

Overall, this project shows that gender does correlate with primary election success, but the relationship isn't straightforward. Instead, the effect of gender depends heavily on the political and structural environment surrounding each candidate. Women tend to perform better in Democratic primaries, while men perform better in Republican primaries, particularly in competitive Senate races. These differences reveal how party culture, recruitment patterns, and voter preferences shape who succeeds. Office type also matters, with Senate races showing lower success rates because they attract more crowded and competitive fields.

Race introduces additional layers, as certain gender–race combinations show stronger or weaker outcomes depending on party context. Our logistic regression confirms that vote share is the dominant predictor of winning, and once we account for vote share and party, gender itself no longer predicts outcomes by itself.

This means that most gender differences come from structural factors: who runs, which offices they run for, and the dynamics of each party, rather than gender acting as a direct barrier or advantage. These findings highlight the importance of studying election outcomes in context rather than in isolation.

The ideology results support this argument by showing how the broader political environment has changed over time. The ideological distance between Democrats and Republicans has grown dramatically since the 1970s, while ideological overlap has nearly disappeared. This long-term polarization helps explain why party differences in primary outcomes are so strong today: candidates operate in a system where the parties are farther apart than ever, leaving little shared ideological space. As polarization increases, party context becomes even more influential in shaping which types of candidates succeed.

Group Contributions: Chiko - Data Cleaning, Visualizations for Question 1/2, Coding Ananya - Report, Coding Melody - Coding, Visualizations for Question 2